Chapter 5

Enthalpy

1. Which one of the following conditions would always result in an increase in the internal energy of a system ____D____?

A) The system loses heat and does work on the surroundings.
B) The system gains heat and does work on the surroundings.
C) The system loses heat and has work done on it by the surroundings.
D) The system gains heat and has work done on it by the surroundings.
E) None of the above is correct.

2. Which one of the following statements is true ___C___?

A) Enthalpy is an intensive property. $\propto$ Mass

B) The enthalpy change for a reaction is ~~in~~dependent of the state of the reactants and products.

C) Enthalpy is a state function. ✓

D) H is the value of q measured under conditions of constant ~~volume~~ Pressure.

$\Delta H = \Delta(E+PV) = \Delta E + P\Delta V = \underbrace{(q+w)}_{\text{First law}} - w$

E) The enthalpy change of a reaction is the ~~reciprocal~~ ตรงข้าม of the ΔH of the reverse reaction.

3. The reaction

$$4Al\ (s) + 3O2\ (g) \rightarrow 2\ Al2O3\ (s) \qquad \Delta H° = -3351\ kJ$$

is ___________, and therefore heat is _____C_____ by the reaction.

A) endothermic, released
B) endothermic, absorbed
C) exothermic, released
D) exothermic, absorbed
E) thermoneutral, neither released nor absorbed

4. Under what condition(s) is the enthalpy change of a process equal to the amount of heat transferred into or out of the system ____B____?

(a) temperature is constant

(b) pressure is constant

(c) volume is constant

$\Delta H = \Delta E + P\Delta V$

$= q + w - w = q$

A) a only
B) b only
C) c only
D) a and b
E) b and c

5. The British thermal unit (Btu) is commonly used in engineering applications. A Btu is defined as the amount of heat required to raise the temperature of 1 lb of water by 1 °F. There are ____C____ joules in one Btu. 1 lb = 453.59 g; °C = (5/9)(°F - 32°); specific heat of H2O(l) = 4.184 J/g-K.

$1\ lb \cdot \frac{453.59\ g}{1\ lb} \cdot 4.184\ J/g{\cdot}K \cdot 1.8\ \frac{°F}{°C}$

=

A) 3415
B) 60.29
C) 1054
D) 5.120 × 10-3
E) Additional information is needed to complete the calculation.

6. A 22.44 g sample of iron absorbs 180.8 J of heat, upon which the temperature of the sample increases from 21.1 °C to 39.0 °C. What is the specific heat of iron ____B____?

A) 0.140
B) 0.450
C) 0.820
D) 0.840
E) 0.900

7. Of the following, ΔH_f° is not zero for ____D____.

A) O_2 (g)
B) C (s, graphite)
C) N_2 (g)
D) F_2 ~~(s)~~ g
E) Cl_2 (g)

8. Consider the following two reactions:

$$A \rightarrow 2B \quad \Delta H°rxn = 456.7 \text{ kJ/mol}$$
$$A \rightarrow C \quad \Delta H°rxn = -22.1 \text{ kJ/mol}$$

Determine the enthalpy change for the process: ____A____

$$2B \rightarrow C$$

A) -478.8 kJ/mol
B) -434.6 kJ/mol
C) 434.6 kJ/mol
D) 478.8 kJ/mol
E) More information is needed to solve the problem.

9. For which one of the following reactions is the value of $\Delta H°_{rxn}$ equal to $\Delta H_f°$ for the product ____A____?

$\Delta H_f°$ reactant = 0

A) $2\ C\ (s,\ graphite) + 2\ H_2\ (g) \rightarrow C_2H_4\ (g)$

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B) $N_2\ (g) + O_2\ (g) \rightarrow 2\ NO\ (g)$

C) $2\ H_2\ (g) + O_2\ (g) \rightarrow 2\ H_2O\ (l)$

D) $2\ H_2\ (g) + O_2\ (g) \rightarrow 2\ H_2O\ (g)$

E) $\cancel{H_2O}\ (l) + 1/2\ O_2\ (g) \rightarrow H_2O_2\ (l)$